

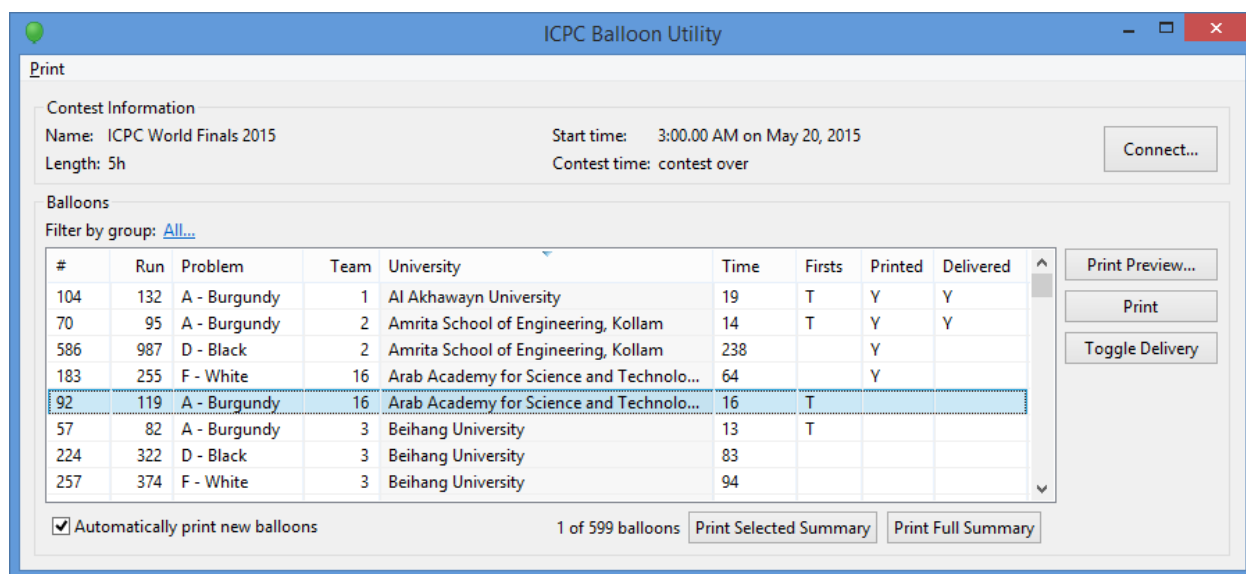
The ICPC Balloon Utility

An ICPC Tool

Introduction

The ICPC Balloon Utility (BU) is a tool for keeping track of balloons earned by contest teams for successful run submissions. It receives submission information from one of several input sources, displays information about each submission (such as the problem, team, run ID, school, submission time, etc.) and optionally automatically prints "balloon notification" messages to assist runners in delivering balloons. It also provides runners with an interactive mechanism for tracking delivered balloons, and is even capable of printing maps showing runners where to go find the appropriate balloon and where on the contest floor the receiving team is located (although instructions for configuring it to do that are beyond the scope of this introductory guide).

The following shows a screen shot of the Balloon Utility main screen, after it has been loaded with run submission information from a Contest Control System:



Using the Balloon Utility

Installation

To install the Balloon Utility, download its distribution package from the [ICPCTools website](#) and unzip it to any convenient location. The BU is a Java program. The distribution is a self-contained package which contains all the Java libraries and other components necessary to run the Balloon Utility. (Note that Java version 17 or higher must also be installed on the machine.)

Execution

The Balloon Utility distribution includes a set of scripts which can be used to launch the program, `balloon.bat` for Windows platforms and `balloon.sh` for macOS and Linux. To start the BU, open a terminal window (command shell), change to the folder where you unzipped the distribution, and type the name of the appropriate script. The scripts do not require any parameters, although adding `--help` will generate a mildly-useful message, and `--clean` will remove any saved balloon data. For more advanced use, you can enter a Contest API URL, user, and password to connect directly to a Contest API server upon startup.

Input Data Sources

When the BU starts it displays a screen similar to the following:

the underlying system and any of these can be selected for use by the BU.

The selected printer can be changed at any time (again, using the Print menu), and the same menu also allows for sending a "Test Message" to the printer to be sure it is working.

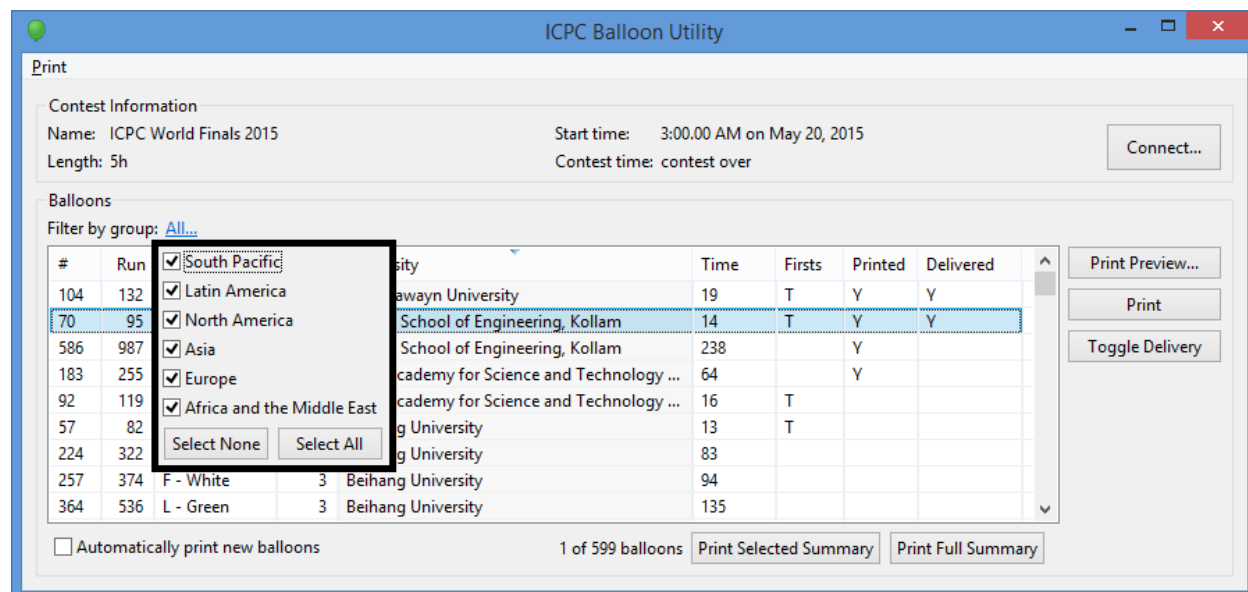
The *Automatically print new balloons* checkbox at the bottom of the main screen can be used to enable or disable automatic printing; if enabled, then whenever a new successful-run notification is received in the event feed the BU will automatically print a new Balloon Notification message to the currently-configured printer.

Selecting one or more rows in the grid (by clicking on them) and then pressing *Print Preview* will print an on-screen version of the Balloon Notification(s) for the selected row(s). Selecting one or more rows and pressing *Print* will send Balloon Notification(s) for the selected rows to the currently configured printer (even if the notifications have previously been printed). Whenever a notification has been printed (including as a result of "automatic printing"), the BU places a "Y" indication in the "Printed" column.

The *Print Summary* buttons allow you to print a complete list of balloons for verification, either a list for rows currently selected in the grid or a complete summary of all balloon notifications for the contest up to the moment. This can be used for example to verify that the balloons which should have been sent to a given team are actually present on the contest floor (by sorting on the Team column and then selecting all runs for that team followed by pressing *Print Selected Summary*), or to check that all of the balloons on the contest floor match what they should be.

The "Delivered" column is used to track when balloons have actually been delivered to teams. Each time a runner delivers a balloon, they should select the corresponding row in the grid and click the *Toggle Delivery* button. This will insert a "Y" in the "Delivered" column, providing an easy way to keep track of delivered notifications. The "Y" notification can be removed if desired by selecting the row and clicking "Toggle Delivery" again. Note also that Balloon Notification printouts have a "Delivered By" signature space near the bottom to allow balloon runners to record that they delivered the corresponding balloon.

The *Filter by group* label (just above the main screen grid, on the left side) can be used to restrict the balloons which are processed by the BU. Clicking on the text next to the label (it shows as "All..." in the following screenshot) produces a drop-down list of the currently-defined "groups" (also called "regions") defined in the contest, similar to the following:



Disabling one or more groups (by unchecking the corresponding checkbox) suppresses the Balloon Utility's processing of Balloon Notifications for the disabled group(s). This allows the balloon handling crew at contests taking place simultaneously at multiple sites to avoid having to deal with Balloon Notifications for teams which are not at their site.

Note that the definition of the groups in a contest comes from data in the event feed read by the BU; groups cannot be defined within the Balloon Utility.

Notes Test or rehearsal contests often have multiple copies of the same problem, e.g. there are 6 problems A-F, but each pair of A and D, B and E, and C and F are identical. If these problems are marked as duplicates by the Contest Control System (by having the same problem UUID) then by default the balloon utility will only print a balloon for first solution (either A or D) from each pair. To disable this behaviour, enable the "Print balloons for problems with matching UUIDs" preference (and restart if there are existing submissions).